

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Second Semester of M. Tech. Mechanical Engineering (AMT) Examination****May 2018****ME 765/ME 746 Surface Engineering****Date: 10.05.2018, Thursday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 (a) Select the appropriate answers.****[04]**

1. Which cannot be altered through surface engineering?

(i) Corrosion resistance.	(ii) Electrical and electronic properties.
(iii) Internal energy.	(iv) Both (i) and (iii).
2. Which of the following statement is not correct regarding sputtering?

(i) The key principle of sputtering is “Energy and Momentum Conservation”.	(ii) Maximum energy transfer occurs in such a collision when the masses are not equal.
(iii) Maximum energy transfer occurs in such a collision when the masses are equal.	(iv) In any collision, momentum is conserved.
3. Which of the following is not the conventional coating method?

(i) Thermal spray.	(ii) Dry coating.
(iii) Compression method.	(iv) Both (i) and (iii).
4. For thermal evaporation process which statement is not true?

(i) A thermal evaporator uses an electric resistance heater to melt the source material.	(ii) Evaporation takes place in open environment.
(iii) Films can also be co-evaporated using multiple sources.	(iv) Evaporated particles can travel directly to the deposition target without colliding with the background gas.

Q-1 (b) Why is surface preparation important?**[01]****Q-1 (c) Differentiate gross cleaning and specific cleaning.****[02]**

- Q-2 (a) Elaborate abrasive surface cleaning processes process. [07]
(b) Categorize specific cleaning processes with applications. [07]

OR

- Q-2 (a) Which are the benefits of surface engineering and where are they used? [07]
(b) Classify surface engineering technology and discuss any one technology with application. [07]

- Q-3 Attempt **any two** [14]
(a) Explain Atomic Force Microscopy (AFM).
(b) Explain in brief substrate surfaces.
(c) Explain out-diffusion. Explain in brief mechanical and thermal properties of surfaces.

SECTION – II

- Q-4 (a) List out the applications of thin film coating in mechanical field. [02]
(b) What are shortfalls of DC sputtering? [02]
(c) What would be the consequences of an interruption in cooling water flow to a magnetron sputter source? [02]
- Q-5 (a) What do you mean by sputtering also explain the RF sputtering process with experimental set up diagram. [07]
(b) Discuss the qualitative study for preparation of thin films by thermal evaporation method. [07]

OR

- Q-5 (a) What is gas transport and how it is useful in CVD systems? [07]
(b) Explain Plasma-Enhanced CVD process and write its advantages. [07]
- Q-6 Write short notes on **any three**. [15]
(a) Pulsed Laser Deposition (PLD).
(b) DC glow discharge with various regions.
(c) Electron Beam Evaporation (EBE)
(d) Ion plating deposition technique.
